

A Slit That Opens Late

How WaveVisualizationModifier lets an interference pattern grow instead of appearing at once

BlenderMathAnim — geometry_nodes/modifier_video_interferences.py

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1 The question

In `towards_double_slit` (`video_interferences/scene_interference.py`) a single slit has been radiating for the whole of the previous scene. At $t_{\text{open}} = 8\text{ s}$ the outer piece of the wall slides aside and a second slit opens. What the surface behind the wall then shows is *not* the finished two-slit pattern. Instead a circular disturbance leaves the new slit, and only the region it has already swept over carries the hyperbolic nodal lines of the interference pattern; outside it the old single-slit wave is untouched. After about seven seconds the disturbance has crossed the whole surface and the stationary pattern is complete.

Every wave this modifier draws is a *steady state*: the field of a source that has been ringing for ever. Such a field is non-zero everywhere at every instant, so a steady-state source that is simply switched on would make the full pattern pop into existence in a single frame. This note explains the few nodes that turn “switched on” into “spreading out”.

2 The steady state

The surface is a grid in the x - y plane whose vertices are lifted by

$$u(\mathbf{r}, t) = \sum_j w_j(\mathbf{r}, t), \quad w_j = A' \left[J_0(kr_j) \cos \omega t + Y_0(kr_j) \sin \omega t \right] = A' \operatorname{Re} \left[H_0^{(1)}(kr_j) e^{-i\omega t} \right], \quad (1)$$

with $r_j = |\mathbf{r} - \mathbf{c}_j|$ (held above `source_radius` to stay clear of the logarithmic pole of Y_0), $k = 2\pi/\lambda$, $\omega = 2\pi f$ and $A' = A\pi/\sqrt{\lambda}$. Each w_j is the exact outgoing solution of the 2+1 dimensional wave

equation for a point source at $\mathbf{c}_j$, and because the equation is linear their sum is a solution too: with two sources it is the two-slit pattern.

The scene builds the surface behind the wall as

```

wave2 = WaveVisualizationModifier(name="DoubleSlitWave", size=8.0,
    sources=((-4, 0), (-4, -a)),
    source_impacts=(None, t_open + 0.5),
    wavelength=1.5, frequency=1.0, amplitude=0.31,
    material=get_texture("function", alpha_intensity=0.9),
    source_radius=2 / 10, time_shift=time_shift,
    intensity=True, intensity_gain=2.0)

```

The grid has side 8 and is moved by +4 in x , so its left edge, local $x = -4$, lies on the wall. Source 0 is the old slit (it slides from $y = 0$ to $y = +a$ before the second one opens); source 1 is the new slit at $y = -a$, $a = 1.25$. The one argument that matters here is `source_impacts`: `None` for the old slit, which rings for ever, and the time $t_1 = t_{\text{open}} + 0.5 = 8.5$ s for the new one — the middle of the one second during which the wall slides open.

3 The causal envelope

A source with an impact time t_j is multiplied by an envelope that is zero ahead of a circle expanding from $\mathbf{c}_j$ at the phase speed $c = \lambda f$, and one behind it:

$$\text{env}_j(\mathbf{r}, t) = S\left(\frac{c(t - t_j) - r_j}{\Delta}\right), \quad S(x) = \hat{x}^2(3 - 2\hat{x}), \quad \hat{x} = \text{clamp}_{[0,1]}(x). \quad (2)$$

- Before the impact $c(t - t_j) < 0$, so $\hat{x} = 0$ at every vertex and the source contributes nothing. The clamp is the whole reason the new slit is silent until it opens.
- After the impact the *reach* $c(t - t_j)$ grows linearly. A vertex at distance r_j starts to feel the source at $t = t_j + r_j/c$ and has the full steady-state contribution a time Δ/c later.
- Δ is the `FrontWidth` dial, one wavelength by default. The smoothstep S has zero slope at both ends, so the edge of the disturbance does not read as a crease in the surface.

The surface therefore shows

$$u = w_0 + \text{env}_1 w_1 = \begin{cases} w_0 & r_1 > c(t - t_1) \quad (\text{single slit, untouched}) \\ w_0 + w_1 & r_1 < c(t - t_1) - \Delta \quad (\text{full two-slit pattern}) \end{cases}$$

with a band of width Δ in between. The disk inside which the pattern is already stationary grows at speed c around the new slit. That is the “slow growth” seen in the shot: the interference is not faded in, it is *carried* outwards by the front.

How long it takes

With $\lambda = 1.5$ and $f = 1$ the front moves at $c = 1.5$ units per second. The farthest vertex from the new slit, local $(-4, -1.25)$, is the corner $(4, 4)$, at distance $\sqrt{8^2 + 5.25^2} \approx 9.6$. The front gets there after 6.4 s, and the band behind it after another $\Delta/c = 1$ s. That is the seven-second camera zoom that follows the opening of the slit.

4 Two clocks: the envelope starts, the phase does not

The crucial design decision is *which* clock each part reads.

phase: $\omega t \rightarrow 2\pi f(t + t_{\text{shift}})$	common to all sources
envelope: $c(t - t_j)$	per source, from its own impact

The phase of the late source is *not* restarted at t_j . Physically the second slit is lit by the same plane wave as the first, whose crests reach the whole wall at once, so the moment the new slit is open it oscillates in step with the old one. Had its phase been measured from t_j instead, the two slits would be out of phase by ωt_j , and the central maximum would move off the symmetry axis.

This is the difference from the other transient the modifier offers, `front=True`, which is a stone dropped into a pond: there the clock is $t - t_{\text{impact}}$ for the phase as well, so the wave leaves its source at zero elongation. The two options are mutually exclusive, and the constructor raises if both are given.

`TimeShift` only enters the phase. It is what lets this scene continue the wave of `towards_slit` without a jump (see `_slit_time_shift`); the impacts and the fronts stay on the scene time of the current scene.

5 The node setup

The tree is three frames. Only the first two know about the late source; the geometry frame is the same as for any other wave.

5.1 Control frame

node (label)	type	role
Clock	Scene Time $\rightarrow$ Seconds	t , no keyframes needed
TimeShift	Value	added to t in the phase only
Wavelength, Frequency	Value	λ , f ; also give $c = \lambda f$
Amplitude	Value	A
Source0, Source1	Vector	$\mathbf{c}_0$, $\mathbf{c}_1$ (animatable)
FrontWidth	Value	Δ , built only if some source has an impact
Impact1	Value	t_1 ; one <code>Impact<j></code> per late source
Intensity, IntensityGain	Value	only with <code>intensity=True</code>

`Impact1` is an ordinary value node, so a scene can still move the moment of opening after the tree has been built: `ibpy.get_geometry_node_from_modifier(wave2, "Impact1")`.

5.2 Wave frame

All of the arithmetic is one `make_function` group named `Elongation`. Its auxiliary expressions, in reverse Polish notation and in the order in which they are emitted, are for the two sources of the scene ($j = 1$ is the late one):

name	RPN	meaning
<code>k</code>	<code>tau,wavelength,/</code>	$2\pi/\lambda$
<code>wt</code>	<code>time,time_shift,+,frequency,*,tau,*,</code>	common phase $\omega(t + t_{\text{shift}})$
<code>amp</code>	<code>amplitude,pi,*,wavelength,sqrt,/</code>	A'
<code>r0, r1</code>	<code>pos,c1,sub,length</code>	r_j
<code>x0, x1</code>	<code>r1,0.2,max,k,*,</code>	$k \max(r_j, r_{\text{src}})$
<code>reach1</code>	<code>wavelength,frequency,*,time,impact1,-,*,</code>	$c(t - t_1)$
<code>gate1</code>	<code>reach1,r1,-,edge/,0,max,1,min</code>	$\hat{x}$
<code>env1</code>	<code>gate1,gate1,*,3,2,gate1,*,-,*,</code>	$S(\hat{x})$
<code>cj_j, sj_j</code>	<code>amp,x1,j0,*,env1,*, (and y0)</code>	gated quadratures
<code>w_j</code>	<code>cj1,wt,cos,*,sj1,wt,sin,*,+</code>	w_j of eq. (1), gated
<code>u</code>	<code>w0,w1,+,</code>	elongation
<code>C, S</code>	<code>cj0,cj1,+, sj0,sj1,+,</code>	for the intensity

The three entries `reach1`, `gate1`, `env1` are produced by `wave_front_gate` in `geometry_nodes/nodes.py`, called as

```
wave_front_gate("r1", "1", "time,impact1,-")
```

— the third argument is exactly where the two clocks part: the gate gets `time,impact1,-`, while `wt` keeps `time,time_shift,+`. Source 0 has no impact, so it gets no gate and its `cj0`, `sj0` carry no `env0` factor.

Two details of the order matter:

- The gate entries are inserted into `aux_functions` *before* the `wj` that multiplies by them; `make_function` emits aux expressions in insertion order, and a name defined later does not exist yet.
- With `intensity=True` the gate multiplies the two quadratures $C_j = A'J_0 \text{env}_j$ and $S_j = A'Y_0 \text{env}_j$ separately, not just their combination. So the time-averaged intensity $C^2 + S^2$, which the `Intensity` dial fades the surface to, is gated in the same way and grows from the new slit just like the wave.

The group has two outputs: `elongation` = $u(1 - I)$ and `value` = $u(1 - I) + gI(C^2 + S^2)$, with I the `Intensity` dial and g the `IntensityGain`. At $I = 0$, the whole of this scene until its last seconds, both are u .

5.3 Geometry frame

Grid → store UVMap → store `result` → store `resultHeight` → store `amplitude` → Set Position → Shade Smooth

The elongation is stored as an attribute while the grid is still flat — a field stored after the lift would measure $\sqrt{r^2 + u^2}$ instead of r — and `Set Position` reads it back through a `Named Attribute`.

The colour needs no separate gating. The "function" material (`function_texture` in `appearance/textures.` paints each point by its `result/amplitude` attribute, i.e. by the number the tree computed, gate included. This is why the scene uses it: the alternative "interference" material recomputes the field in the shader and knows about `front`, but not about `source_impacts`, so it would paint the finished pattern on a surface that has not yet heard of the second slit.

6 A look, not a solution

Equation (2) is not what the wave equation does. The true field of a source switched on at t_1 is the retarded convolution with the 2D Green's function $\propto \Theta(c\tau - r)/\sqrt{c^2\tau^2 - r^2}$, which has a sharp front at $r = c(t - t_1)$ but, unlike in three dimensions, a wake behind it: it approaches the steady state only slowly, with a tail decaying like a power of t .

What the envelope gets right are the two things an audience actually reads: nothing changes at distance r from the new slit before $t_1 + r/c$, and behind the front the pattern is the stationary one. A slower, exact relaxation would spend most of the shot on a transient nobody is meant to look at.

7 Checking it without Blender

Every field built into the tree also exists in numpy. `WaveVisualizationModifier.elongation_numpy(points, seconds)` evaluates eq. (1) with `scipy.special.j0/y0` and applies the same gate per source in `source_impacts`, so the build-up can be plotted at, say, $t = 9, 11, 13, 15$ s and compared with frames of the render. (It ignores `Intensity`; at $I = 0$ that is the whole of what the tree computes.)